

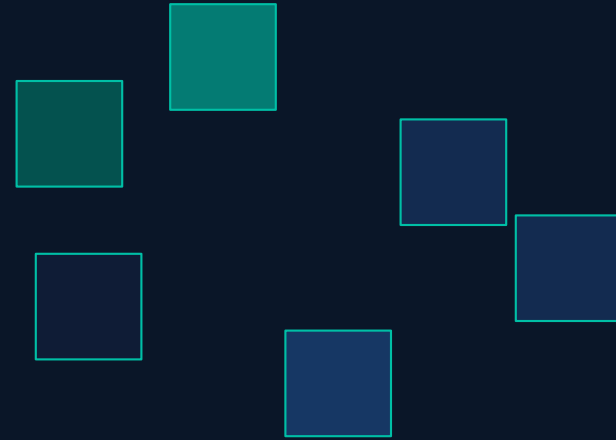


BLOCKCHAIN, DECENTRALIZATION & WEB3

From Cryptocurrency to Real-World Business Applications

Instructor: Dr. Ing Chukwuka Victor Obionwu | **Course:** CSC 322 | **Date:** Week 7

Week 7 — Business & Technology Series



By the End of This Lecture You Will...

- 1 Define blockchain** — Explain what a blockchain is and how its four core properties work
- 2 Understand decentralisation** — Describe the shift from centralised to decentralised systems and why it matters
- 3 Explain Smart Contracts & DAOs** — Articulate how automated contracts and new organisational models work
- 4 Map real-world applications** — Identify blockchain use cases in supply chain, finance, and digital identity
- 5 Evaluate critically** — Assess the genuine opportunities and risks blockchain presents for business

Today's Agenda

01

Foundations

What blockchain is & how it works

18 min

02

Smart Contracts & DAOs

Code as contract, new org structures

12 min

03

Real-World Applications

Supply chain, identity, finance

14 min

04

Risks & Ethics

Challenges every business leader must know

8 min

05

Summary & Q&A

Key takeaways + your questions

8 min

The Core Problem: Who Do You Trust?



Send Money Abroad

Bank takes 3–5 days & charges 5–7% fees. You trust the bank to move YOUR money.



Track Your Parcel

Package goes through 8 hands. Who entered the wrong data? Who can you hold accountable?

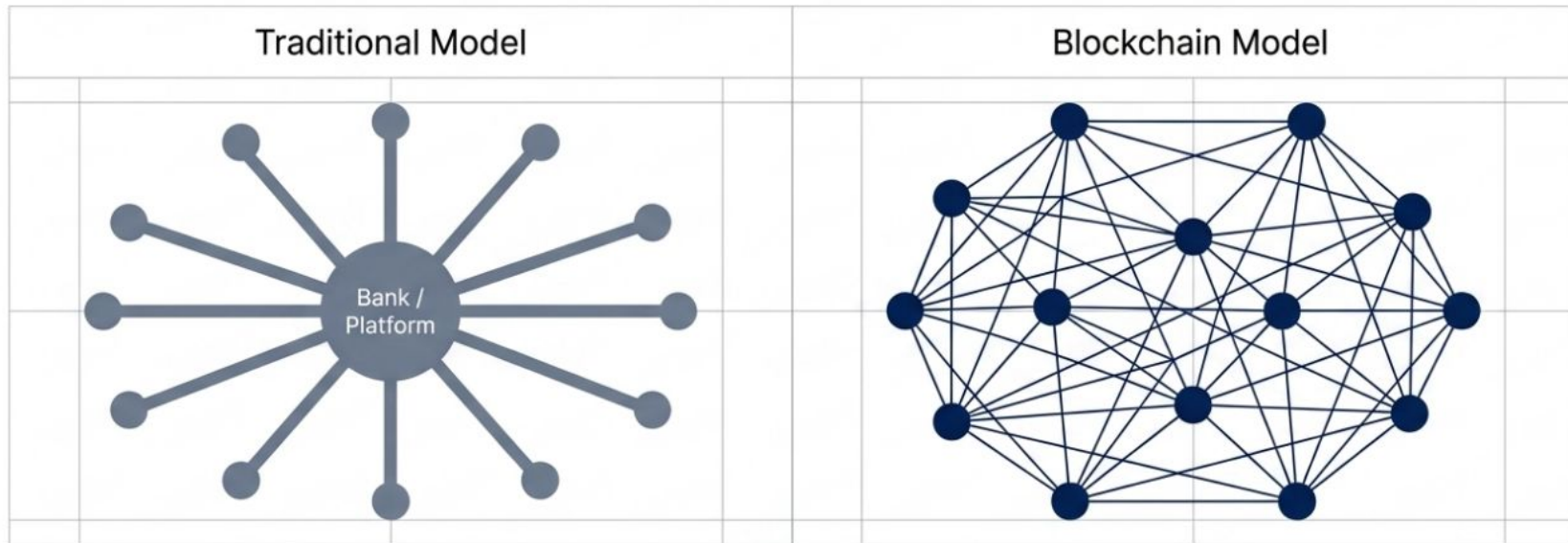


Prove Your Identity

You own 20+ logins. One breach exposes everything. The platform owns your data — not you.

All three share the same root problem: you must trust a centralised middleman.

Solving the Coordination Problem



Traditional: High friction. Single point of failure. Relies on institutional trust.

Blockchain: Cryptographic verification. Distributed consensus. Relies on code-based trust.

Blockchain does not eliminate trust; it automates and redistributes it through mathematics.

What Is a Blockchain?

"A shared, digital ledger that records transactions — permanently, transparently, and without a central authority."



Distributed

Thousands of computers hold identical copies. No single point of failure.



Immutable

Once written, data cannot be altered or deleted without detection.



Transparent

All participants can view the full transaction history in real time.



Decentralised

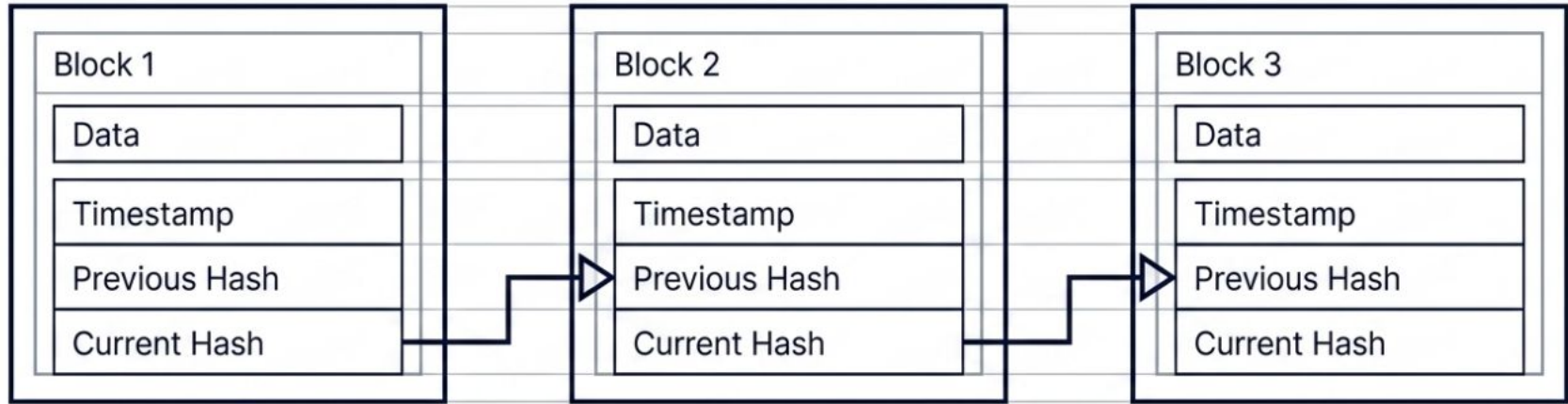
No bank, government, or company controls it — the network does.

How Does a Blockchain Actually Work?



 **CONSENSUS:** Before a new block is added, the majority of the network must agree it's valid. Two main methods: **Proof of Work** (energy-intensive, Bitcoin) vs. **Proof of Stake** (energy-efficient, Ethereum since 2022).

How Does a Blockchain Actually Work?

**Hashing:**

A tiny data change generates a completely new cryptographic fingerprint.

Chaining:

Blocks lock together. Tampering with one breaks the entire sequence.

Consensus:

Network nodes cross-verify chains automatically.

Is Blockchain the Right Tool?

Dimension	Traditional Database	Blockchain Network
Control	Centralized Administrator	Decentralized Consensus
Data State	Easily Editable (CRUD)	Immutable (Append-Only)
Performance	Extremely Fast & Scalable	Slower (Trade-off for security)
Ideal Use	Single-entity environments	Multi-party, low-trust environments



PAIR DISCUSSION — 3 MINUTES

*"Would you trust a contract enforced by code instead of a lawyer?
Why — or why not?"*

Think about:

- What happens if the code has a bug?
- Who resolves disputes?
- What industries could benefit most?

Turn to your neighbour — discuss for 3 minutes, then we'll hear 2–3 responses.

Smart Contracts: Code as a Contract

What is it?

A self-executing program stored on the blockchain. When pre-defined conditions are met, it automatically performs an action — no middleman needed.



Think of a vending machine:

Insert correct amount → product released. No cashier needed. No negotiation. The machine enforces the deal.

IF conditions met → **THEN** action
executes automatically



Crop Insurance

Smart contract pays a farmer automatically when satellite data confirms a drought. No claims form, no adjuster.



Real Estate Escrow

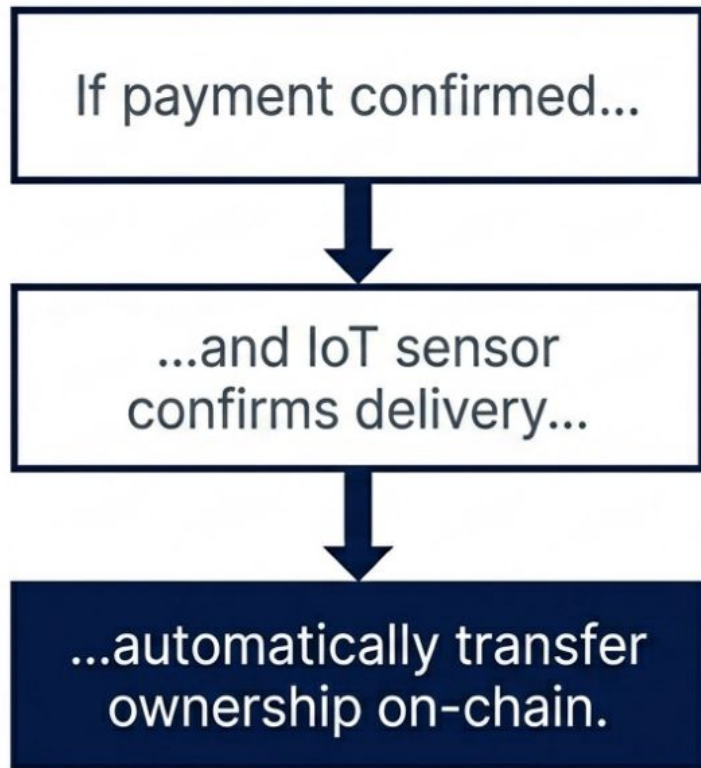
Funds held in smart contract; release triggers when legal checks clear. Cuts weeks off property transactions.



Music Royalties

Every stream triggers an instant micro-payment to the artist. No record label as gatekeeper.

Smart Contracts: Self-Executing Code



Concept: Code written directly onto the blockchain.

Function: Removes human intermediaries for routine agreements.

Risk: "Code is Law" — vulnerabilities or bugs execute without human intervention or safety nets.

Cryptocurrency & Decentralised Finance (DeFi)

\$57B

Global blockchain
market in 2025

**Projected: \$1.4 Trillion
by 2030**

*617M+ crypto users
worldwide (2025)*

Cryptocurrency

Digital money secured by cryptography. Bitcoin: peer-to-peer payments without banks. Ether: fuel for the Ethereum network.

DeFi (Decentralised Finance)

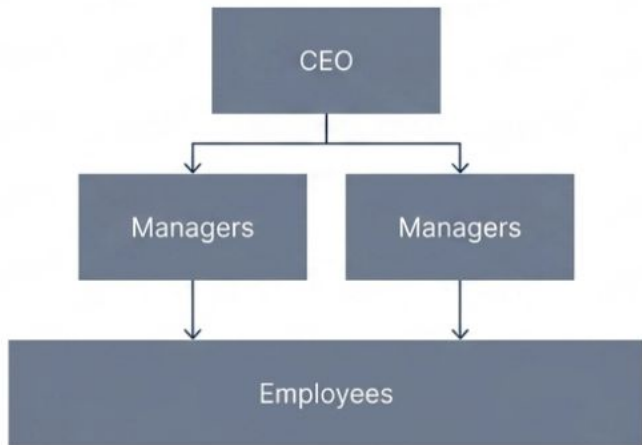
Banking services — lending, borrowing, trading — with no bank. Protocols like Aave let users earn interest or borrow against crypto collateral 24/7.

What it's NOT

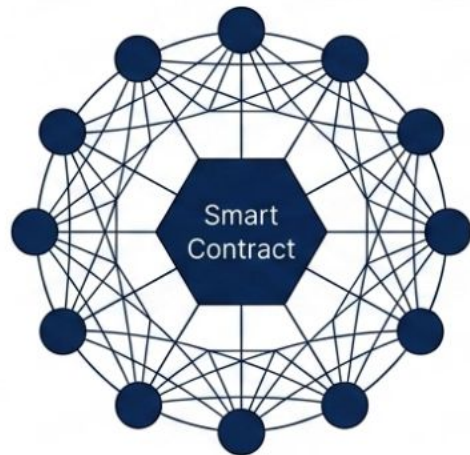
Not anonymous (pseudonymous). Not unregulated forever. Not just speculation — institutional banks (JP Morgan, Goldman Sachs) actively use blockchain for settlement.

Decentralized Autonomous Organizations (DAOs)

Traditional Corporate Pyramid



DAO Circular Node Network



Mechanism

Rules and treasuries encoded in smart contracts; token-weighted voting.

Benefit

Global, permissionless, fully transparent operations.

Challenge

Voter apathy, security risks, and evolving legal status.

DAOs: A New Way to Organise?



💬 *Think of a DAO as "a group chat with a shared bank account and rules written in code."*

How it works

- Members hold governance tokens
- Proposals submitted by anyone
- Token holders vote on decisions
- Smart contracts auto-execute the result

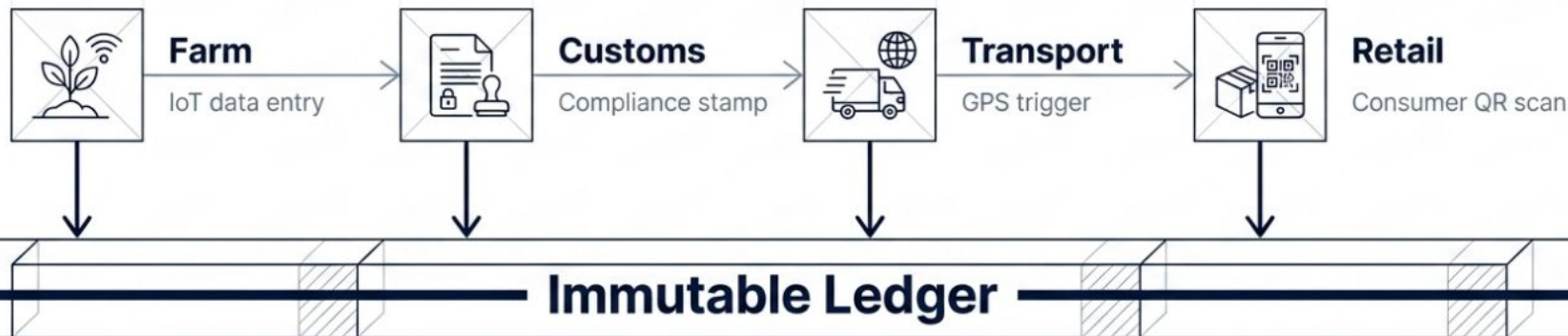
Traditional Company

- CEO + board make decisions
- Shareholders vote once a year
- Lawyers & admin execute changes
- HQ controls the money

Real Examples

- MakerDAO — \$8B+ in assets, governs a stablecoin
- Wyoming became 1st US state to legally recognise DAOs (2021)
- ConstitutionDAO raised \$47M in 72 hours to buy a rare US Constitution copy

End-to-End Traceability



The Problem:

Opaque global supply chains breed fraud, counterfeits, and inefficiency.

The Solution:

Immutable audit trails visible to all stakeholders simultaneously.

Current Status:

20–30% measurable efficiency gains in enterprise deployments.

Case Study: Walmart & IBM Food Trust

The Problem

2006 E. coli outbreak in US spinach: FDA took 2 weeks to trace the source. Entire industry shut down. Lives lost.

The Blockchain Solution

Walmart partnered with IBM to put its entire food supply chain on a blockchain (Hyperledger Fabric). Every farm, distributor, and store uploads data.

*500+ food suppliers
worldwide enrolled*

7 days

Time to trace mango
origin (before)

2.2 sec

Time to trace mango
origin (after blockchain)

80%

*"Our customers deserve a more transparent supply chain." —
Walmart VP, Food Safety*

Supply Chain: Beyond Food



Luxury Goods — LVMH

AURA blockchain consortium tracks every Louis Vuitton bag from raw material to sale. Customers scan a QR code to verify authenticity. Counterfeit luxury goods are a \$450B problem.



Diamonds — Everledger

Each diamond given a digital passport stored on blockchain — cut, clarity, origin. Prevents conflict diamond trade. Over 2 million diamonds registered.



Pharma — Drug Traceability

US Drug Supply Chain Security Act (DSCSA) requires full traceability by 2025. Pfizer and MSD use blockchain to verify drugs aren't counterfeit — critical in markets with 10%+ fake medicines.



Shipping — Maersk / TradeLens

World's largest shipping company built a blockchain platform handling 60% of global container shipping data. Cut documentation time from days to hours.

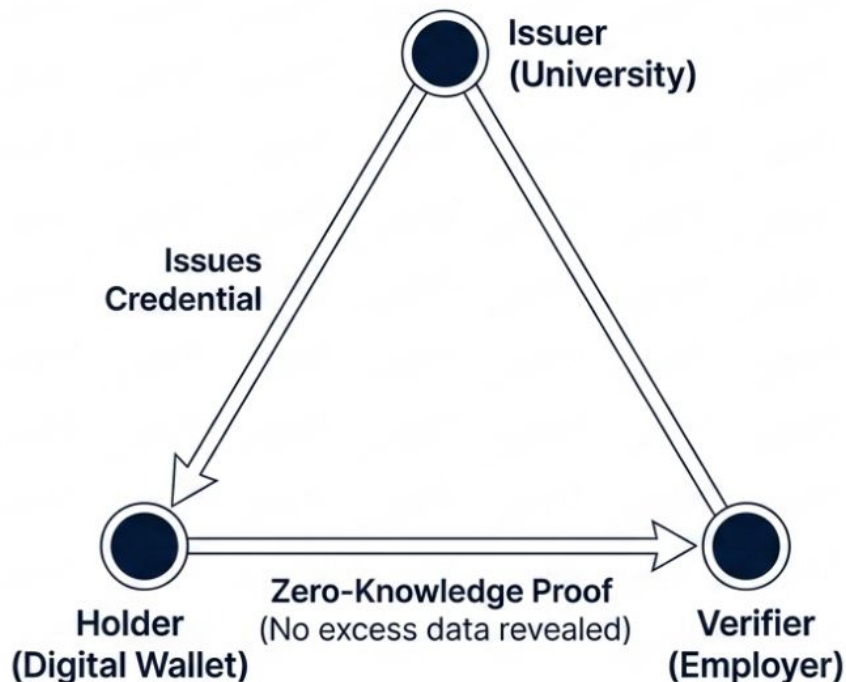
Blockchain & Digital Identity



Self-Sovereign Identity (SSI): YOUR credentials (degree, passport, ID) stored in a digital wallet YOU control — not Facebook, not a bank, not a government server. Share only what's needed. Verifiable. Portable.

Real-world: EU Digital Identity Wallet (eIDAS 2.0) launching across all 27 EU member states — 450M people gaining blockchain-compatible digital IDs.

Owning Your Digital Identity



Privacy by Design:

Users control what data they share.

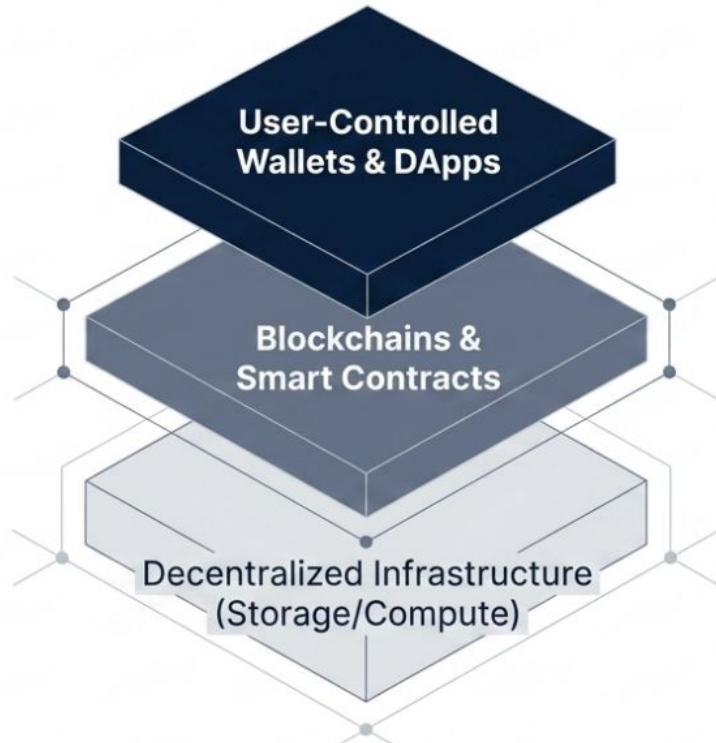
Portability:

Credentials move with the user, not locked in corporate silos.

Inclusion:

Provides verifiable identity for the ~800 million people lacking official state ID infrastructure.

The Web3 Architecture



Web2 Paradigm: Platforms own the infrastructure, extract user data, and capture the network value.

Web3 Paradigm: Users own the data, control the assets, and govern the protocols via cryptographic wallets.

Web3: A New Internet — or Overhyped?

What Web3 promises:

- Own your digital assets (NFTs, in-game items, art)
- Creator economy: artists paid directly, no platform cut
- Tokenised ownership — own a fraction of a building, artwork, or company
- Open finance available to anyone with internet



NFTs: A digital file + proof of ownership on the blockchain. Beeple sold digital art for \$69M in 2021. Kings of Leon released an album as an NFT. Utility NFTs now used for event tickets, loyalty programmes, and property deeds.



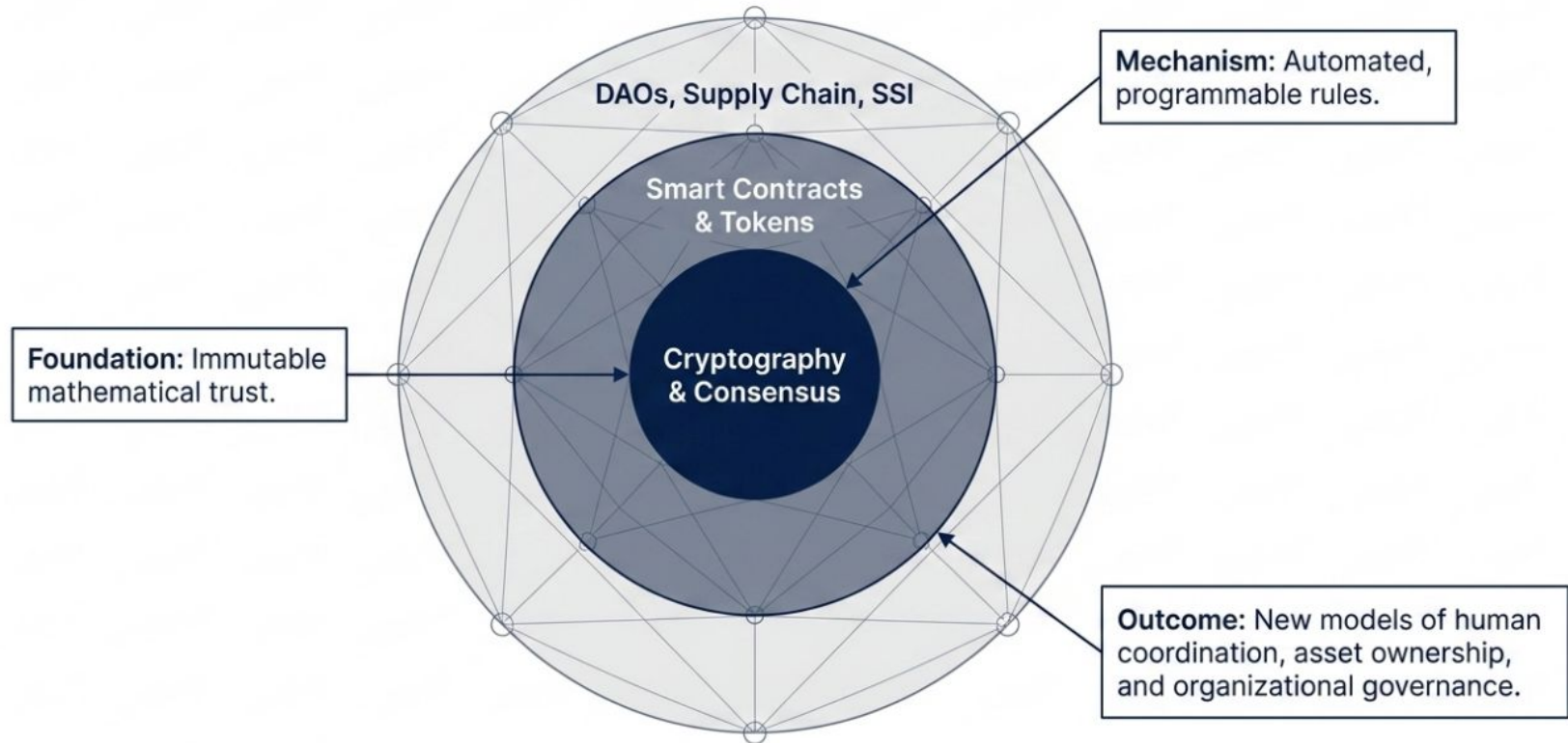
Reality check:

- NFT market crashed 97% from 2022 peak — many were pure speculation
- Most Web3 apps still rely on centralised infrastructure (AWS, Cloudflare)
- User experience is poor — crypto wallets are complex
- Significant environmental concerns (though Ethereum now uses 99% less energy after 2022 upgrade)



Bottom line: Web3 represents genuine shifts in ownership and finance, but current user experience and infrastructure lag far behind the vision. Watch this space — but critically.

The Decentralization Engine





GROUP ACTIVITY — 4 MINUTES

In groups of 3–4, choose ONE sector. Identify: (1) a blockchain use case + (2) one challenge



Healthcare

Hint ideas:

- Patient records / consent
- Drug traceability
- Clinical trial data



Retail & Fashion

Hint ideas:

- Counterfeit prevention
- Loyalty tokens
- Ethical sourcing proof



Music Industry

Hint ideas:

- Royalty micro-payments
- NFT albums
- Ticket fraud

After 4 minutes: one spokesperson per group shares their use case AND their biggest challenge.



The challenge is just as important as the use case — blockchain isn't magic.

Risks, Challenges & Ethical Questions



The Oracle Problem

Blockchain can't verify real-world data at source. Garbage in, garbage out — immutably.



Scalability

Bitcoin processes ~7 transactions/sec. Visa: 24,000/sec. Blockchain at scale remains an engineering challenge.



Fraud & Scams

\$17 billion lost to crypto fraud in 2025. Irreversibility means victims have no recourse — no 'undo' button.



Regulation

Laws vary wildly by country. MiCA regulation in the EU (2024) brings order; US regulation still fragmented.



Energy Use

Bitcoin uses ~150 TWh/year — more than Argentina. Ethereum cut its energy use 99.95% with Proof of Stake in 2022.



Access & Inequality

Web3 may favour those with technical skills and capital. Risk of deepening the digital divide.

Key Takeaways

1

Blockchain solves the trust problem — A shared, immutable ledger removes the need for costly, slow middlemen in transactions.

2

Smart contracts automate business logic — Self-executing code that triggers actions when conditions are met — no lawyers required for routine deals.

3

Supply chain is the #1 enterprise use case — From Walmart's mangoes to LVMH handbags — traceability and transparency are transforming global trade.

4

Web3 shifts ownership to individuals — Identity, assets, and finance are all moving toward models where YOU control your data — not platforms.

5

Scepticism is healthy — Blockchain has real limitations: the oracle problem, scalability, fraud, and access inequality demand critical thinking.



Leave-with question: "In 10 years, which industries on today's slides will look most different — and why?"

References & Further Reading

Academic

1. Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System. bitcoin.org
2. IBM Food Trust case study: ScienceDirect (2025). Tracking transparency: Blockchain in the global supply chain.
3. Yiannas, F. (2018). A New Era of Food Transparency Powered by Blockchain. *Innovations Journal*, 12(1–2).
4. ResearchGate (2021). Food Traceability on Blockchain: Walmart's Pork and Mango Pilots with IBM.

Reports & Industry

1. World Economic Forum (2019). Walmart is betting on blockchain to improve food safety. weforum.org
2. Grand View Research (2025). Blockchain Technology Market Size & Forecast Report.
3. DemandSage (2026). Blockchain Adoption Statistics: Growth, Users, and Market Trends.
4. EU Commission (2024). eIDAS 2.0 — European Digital Identity Wallet Regulation.



Further Reading: Don Tapscott & Alex Tapscott — *Blockchain Revolution* (2016) · Andreas Antonopoulos — *Mastering Bitcoin* (free online, 3rd ed.) · [Ethereum.org](https://ethereum.org) — Official documentation and learning resources

Thank You

Questions & Discussion

*"The technology most likely to change the next decade
is not social media or AI — it is blockchain." — Don Tapscott*

[Instructor Name]

[Email Address] · [Office Hours / Course Platform]

Next week: [Topic Placeholder]